

JavaScript Notes — Day 2

Chapter 2: Operators and Conditional Statements

Today's Topics

- 1 Operators
- 2 Conditional Statements

Multi-line Comments

Use `/* */` to write comments spanning multiple lines, instead of `//` on every line.

```
/* This is a long comment
   that spans multiple lines */
```

1. Arithmetic Operators

Used to perform mathematical operations on values and variables.

Operator	Name	Example	Result
+	Addition	a + b	15
-	Subtraction	a - b	5
*	Multiplication	a * b	50
/	Division	a / b	2
%	Modulus	a % b	0
**	Exponentiation	a ** b	100000
++	Increment	c++	c + 1
--	Decrement	d--	d - 1

Given: a = 10, b = 5. Note: ++ and -- are unary operators (operate on a single operand).

```
let a = 10;
let b = 5;
let sum = a + b; // 15
let difference = a - b; // 5
let product = a * b; // 50
let quotient = a / b; // 2
let remainder = a % b; // 0
let power = a ** b; // 100000
```

2. Assignment Operators

Used to assign values to variables, optionally combined with an operation.

Operator	Meaning	Example (x=10)	Result
=	Assign	x = 10	10
+=	Add & assign	x += 5	15
-=	Subtract & assign	x -= 5	10
*=	Multiply & assign	x *= 5	50
/=	Divide & assign	x /= 5	10
%=	Modulus & assign	x %= 3	1
**=	Exponent & assign	x **= 2	1

3. Comparison Operators

Compare two values and return a boolean (true/false).

Operator	Meaning	Example (e=10, f=10, g="10")	Result
==	Equal to (loose)	e == f	true
!=	Not equal to (loose)	e != f	false
===	Strict equal (value + type)	e === g	false
!==	Strict not equal	e !== g	true
>	Greater than	e > f	false
<	Less than	e < f	false
>=	Greater than or equal	e >= f	true
<=	Less than or equal	e <= f	true

Note: === and !== check both value and type — e === g is false because e is a number and g is a string.

4. Logical Operators

Combine multiple boolean expressions and return a boolean result.

Logical AND (&&): returns true only if both operands are true.

```
let a = 6;
let b = 7;
console.log(a > 5 && b > 5); // true
```

Conditional Statements

if statement

Executes a block of code if a specified condition is true.

```
let age = 18;
if (age >= 18) {
  console.log("You are an adult.");
}
```

if...else statement

Runs one block if the condition is true, another if it's false.

```
let mode = "dark";
let color;
if (mode === "dark") {
  color = "black";
} else {
  color = "white";
}
```

else if statement

Tests a new condition when the previous condition is false.

```
let score = 85;
if (score >= 90) {
  console.log("Grade: A");
} else if (score >= 80) {
  console.log("Grade: B");
} else if (score >= 70) {
  console.log("Grade: C");
} else if (score >= 60) {
  console.log("Grade: D");
} else {
  console.log("Grade: F");
}
```

Ternary Operator

A shorthand for if-else. Takes a condition, a true-case value, and a false-case value.

```
let isMember = true;
let discount = isMember ? 0.1 : 0; // 0.1
```

switch statement

Performs different actions depending on which case matches the value.

```
let day = "Monday";
switch (day) {
  case "Monday":
    console.log("Today is Monday.");
    break;
}
```

```
// ... other days
default:
  console.log("Invalid day.");
}
```

Practice Problem

Take user input for a number and check if it is divisible by 5.

```
let n = prompt("enter a number");
if (n % 5 === 0) {
  console.log(n + " is divisible by 5.");
} else {
  console.log(n + " is not divisible by 5.");
}
```